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Properties of Materials
Lecture 9

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Lecture 6: The Elastic Moduli, Deformation of Brittle Materials, Deformation of Ductile Materials, Hooke's Law - Young's Modulus, Linear Elastic Behaviour, Structural Engineering, What Might Have Saved the Bridge, Engineering Stress and Strain, True Stress and Strain, Comparison
Lecture 7: Practice Time, Definitions, Concepts and Terminology (3), Task
Exercise Seminar 1
Lecture 8: Bonding Between Atoms, Learning Objectives, Bonding Energies, Primary Bonding, Ionic Bonding, Covalent Bonding, Metallic Bonding, Crystal Structures, Body-Centred Cubic, Face-Centred Cubic, Secondary Bonding, Van der Waals Forces, Hydrogen Bonding, Bonding Energies and Properties, Bonding Types and Properties, Concepts and Terminology (4)
Lecture 9: Recap from Last Time, Task

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Bonding between atoms

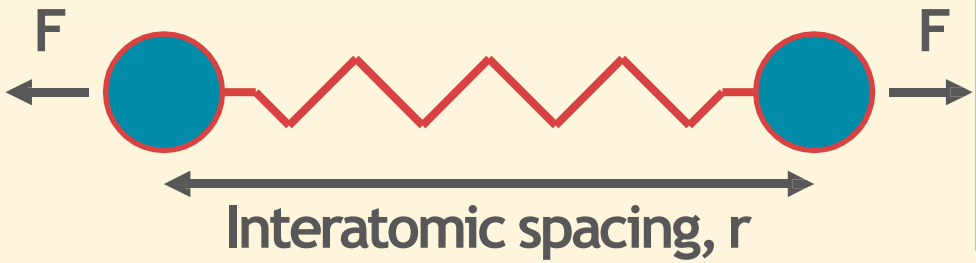
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Recap from last time...

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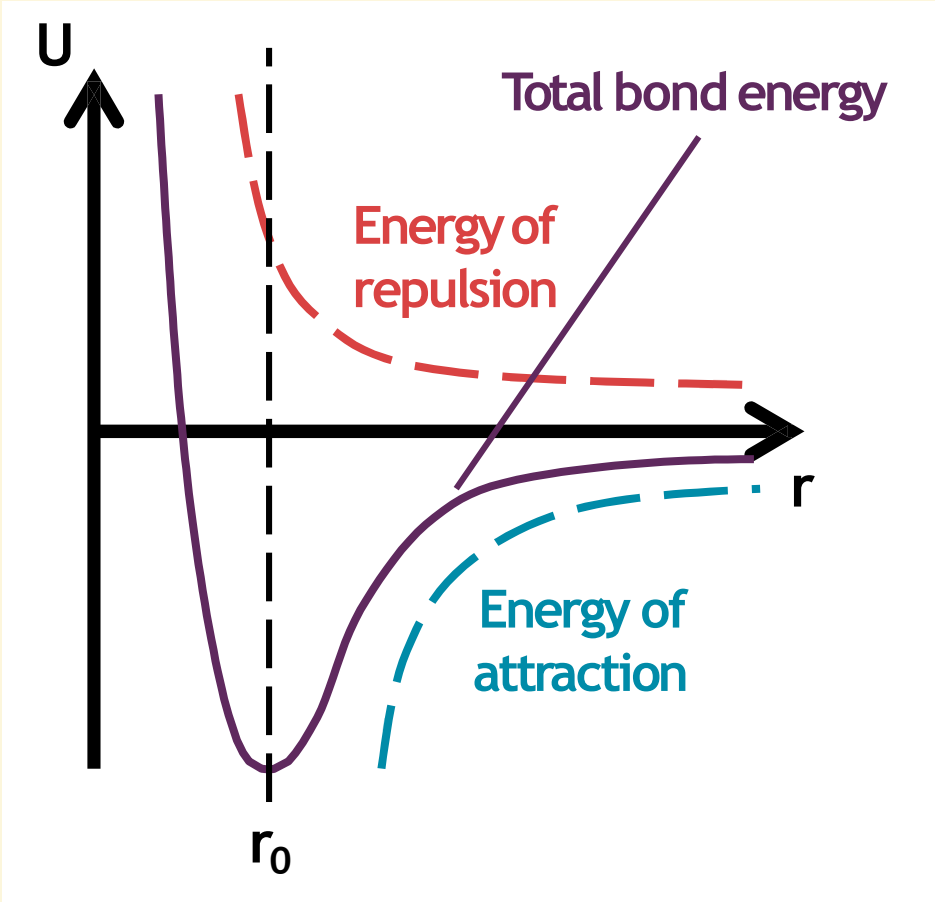
Bonding energies (1)

- Ball and spring model:
  - Atoms subjected to competing repulsive and attractive forces



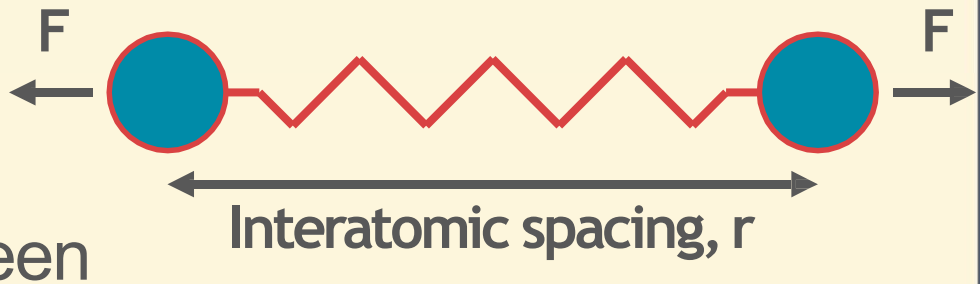
- At rest, atoms will be in their lowest energy state

- r_0 = equilibrium interatomic spacing
  U = potential energy of bond

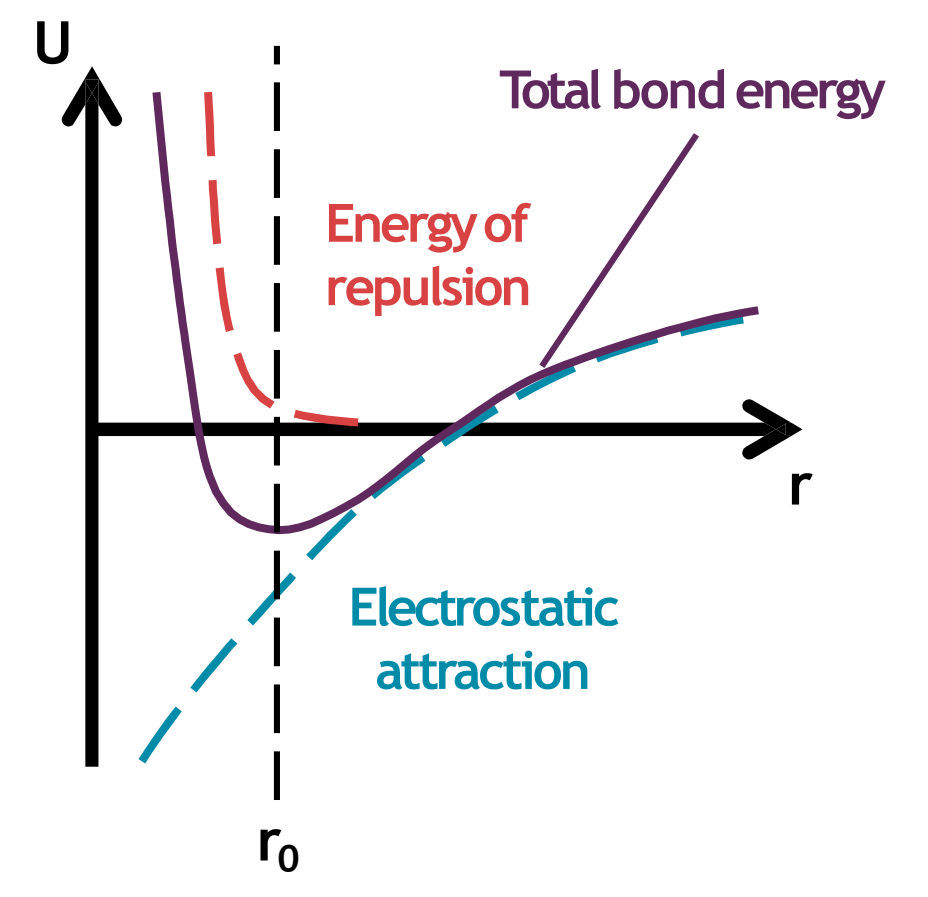


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Ionic bonding (1)

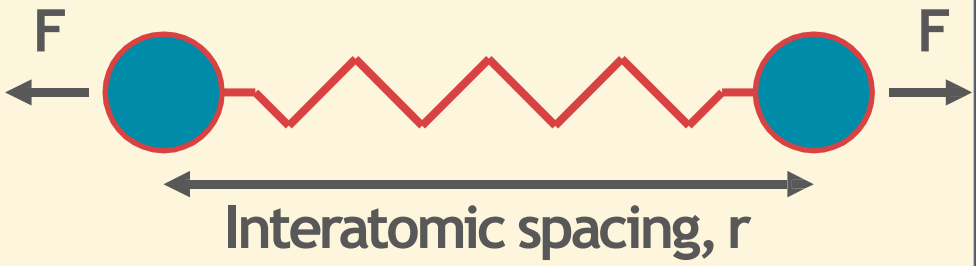


- **Electrostatic attraction** between ions:
  - e.g. $Na^+ + Cl^- \rightarrow NaCl$
- At equilibrium spacing, electrostatic attraction balanced with **repulsion energy**:
  - Ions repulse when electron charge distributions start to overlap
- Ionic bonds are **non-directional** but ionic structures must alternate from +ve to -ve

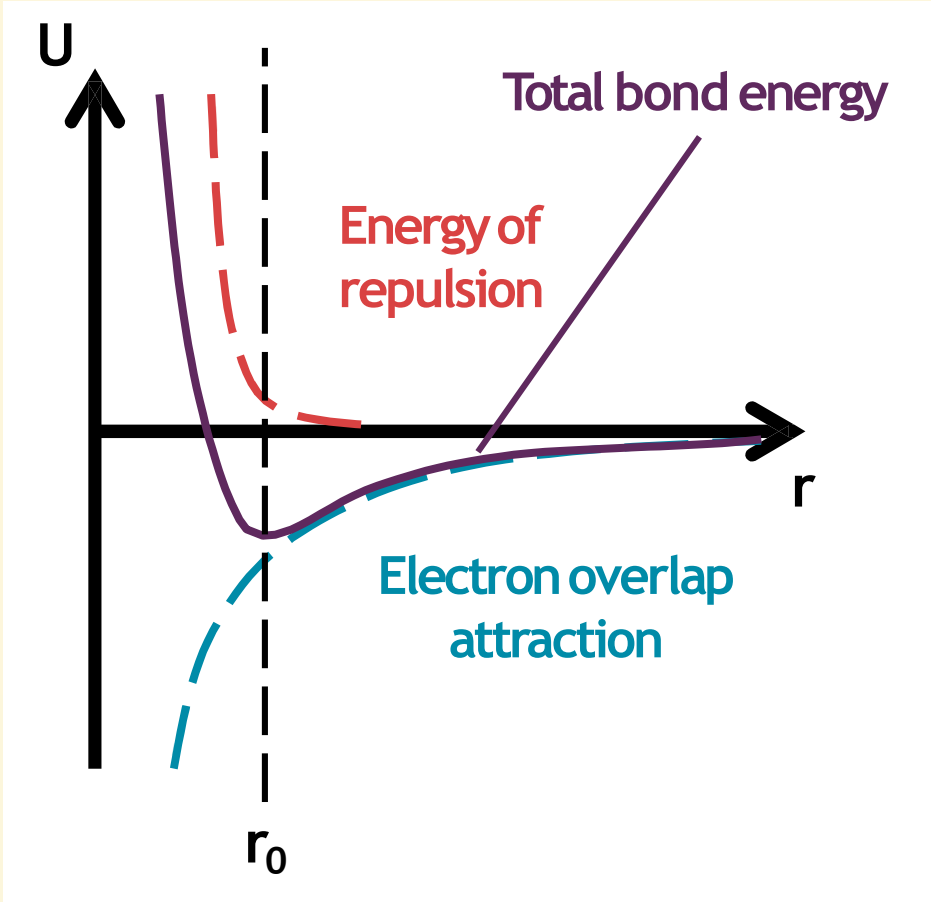


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Covalent bonding



- **Sharing of electron orbitals**:
  - e.g. Diamond, silicon carbide
  - Contributes to bonding of high-melting point metals (e.g. W, Mo, Ta)
  - Bonding in hydrocarbon chains



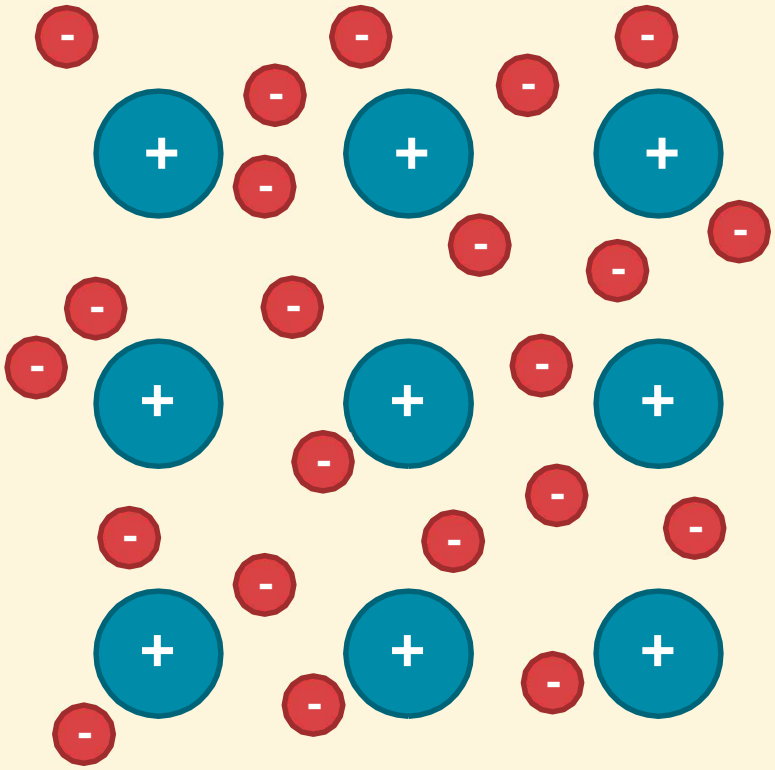
- Covalent bonds are **highly directional**:
  - Influences the way in which atoms 'stack' together to form larger structures

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Metallic bonding

- **Mobile electrons** around fixed metal cations:
  - e.g. Metals!
  - Mobile electrons give the highest electrical conductivity

- Metallic bonds are completely **non-directional**:
  - Atoms can 'stack' close together while electrons flow around them

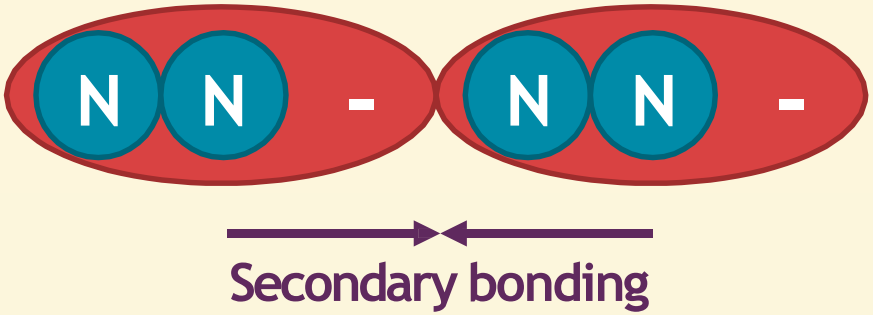


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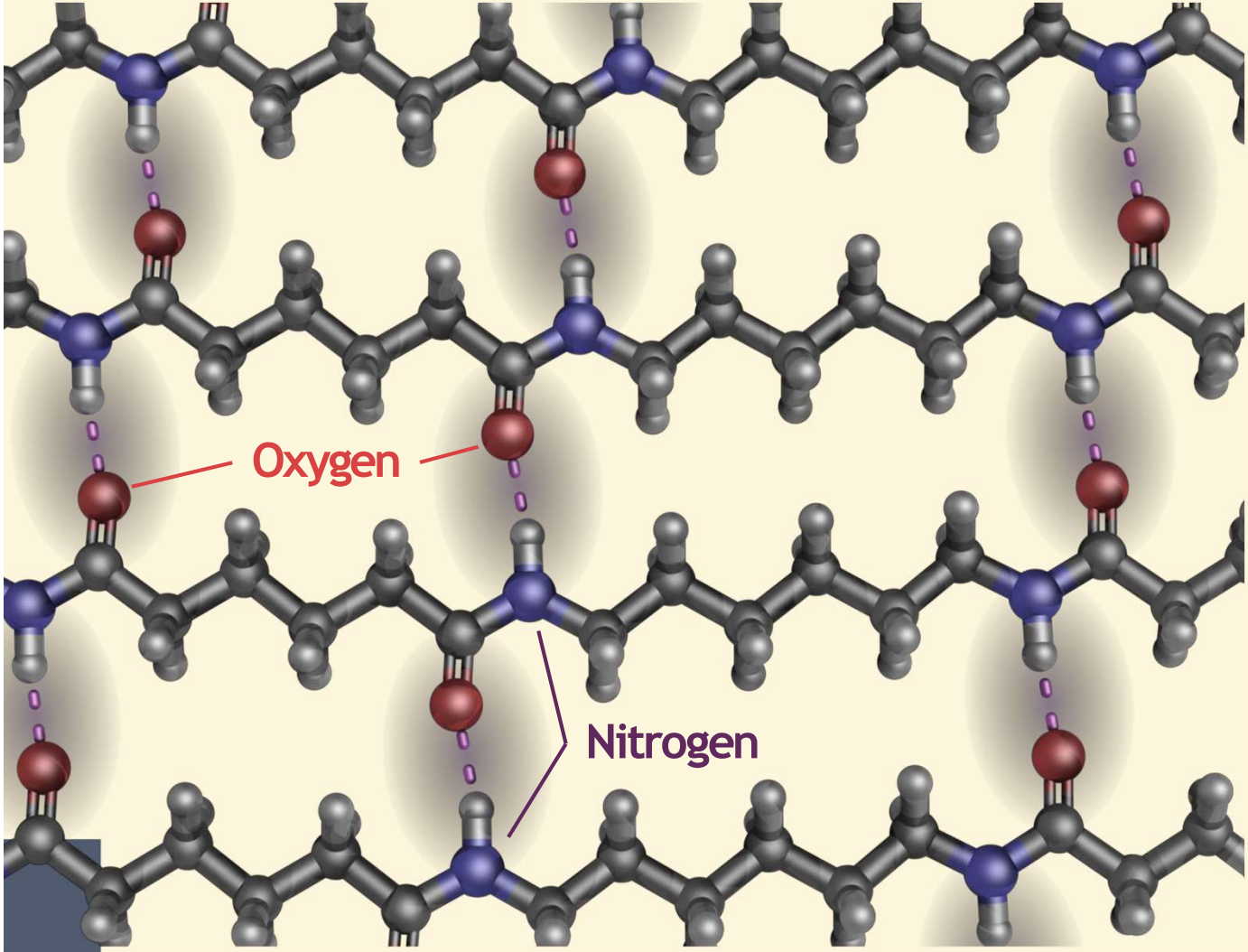
Van der Waals forces

- **Dipolar attraction** between uncharged atoms
  - Electron movement causes **fluctuation in charge** on individual atoms, which then interact to create dipoles on others

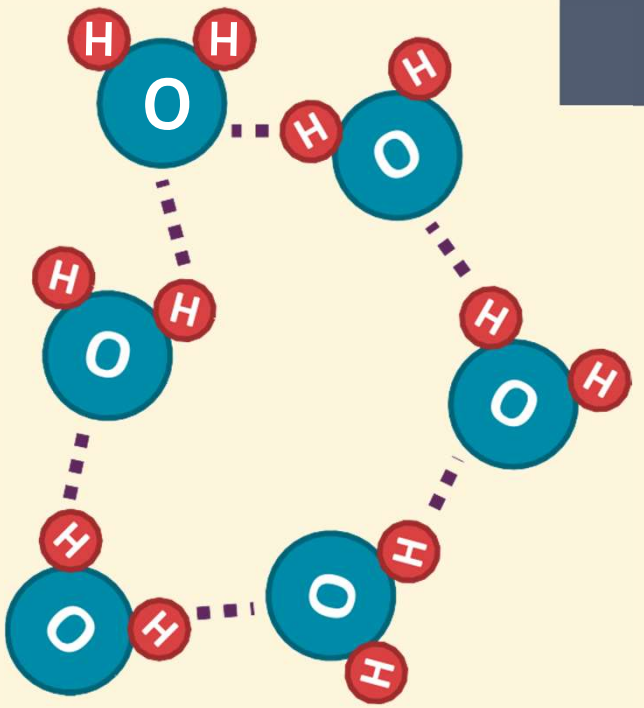
- Without Van der Waals forces, **liquefaction** would not be possible for many important industrial liquids:
  - Liquid nitrogen, oxygen, helium and hydrogen



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Hydrogen bonding in nylon6,6
https://commons.wikimedia.org/wiki/File:Nylon-3D-h_bond.png



Hydrogen bonding

- Hydrogen atoms are able to **share their charge**:
  - Only with nitrogen, oxygen or fluorine in molecules
- Essential for solid polymers and keeping water liquid!

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Bonding types and properties

- **Ceramics**:
  - Large bond energy
  - Stiff and strong, but brittle
  - High melting point and low thermal expansion coefficient
  - **Ionic** and **covalent** bonding
- **Metals**:
  - Variable bond energy
  - Moderate stiffness and strength, can be brittle or ductile
  - Tend to have high thermal and electrical conductivity
  - **Metallic** bonding
- **Polymers**:
  - Directional properties
  - Low stiffness and strength; poor thermal properties
  - **Covalent** and **secondary** bonding

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Task!

- Use internet to find out the nature of bonding between carbon fiber-epoxy matrix reinforced composite!